



SUPER STORE SALES PERFORMANCE ANALYSIS REPORT



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1. Executive Summary

This report provides a comprehensive analysis of the SuperStore sales data from 2014 to 2017. The primary goal was to identify key drivers of profitability and uncover actionable insights to improve business performance. The analysis revealed three critical findings: (1) The company's financial health is heavily imbalanced, with the West region being highly profitable while states like Texas and Pennsylvania are major loss centers. (2) A strong seasonal pattern exists, with sales consistently peaking in the final quarter (Q4) of each year. (3) High discounts are directly correlated with significant profit losses. Key recommendations include a strategic review of underperforming regions, optimizing inventory for the Q4 peak, and implementing a data-driven discount policy to protect profit margins.

2. Introduction

The objective of this project was to conduct an in-depth exploratory data analysis (EDA) of the SuperStore dataset. The goal was to move beyond surface-level metrics and understand the complex relationships between geography, product categories, shipping methods, and profitability. This report aims to answer critical business questions and provide data-backed recommendations to drive strategic decisions.

3. Data Preparation

Before analysis, the raw data was processed to ensure accuracy and reliability. The key preparation steps included:

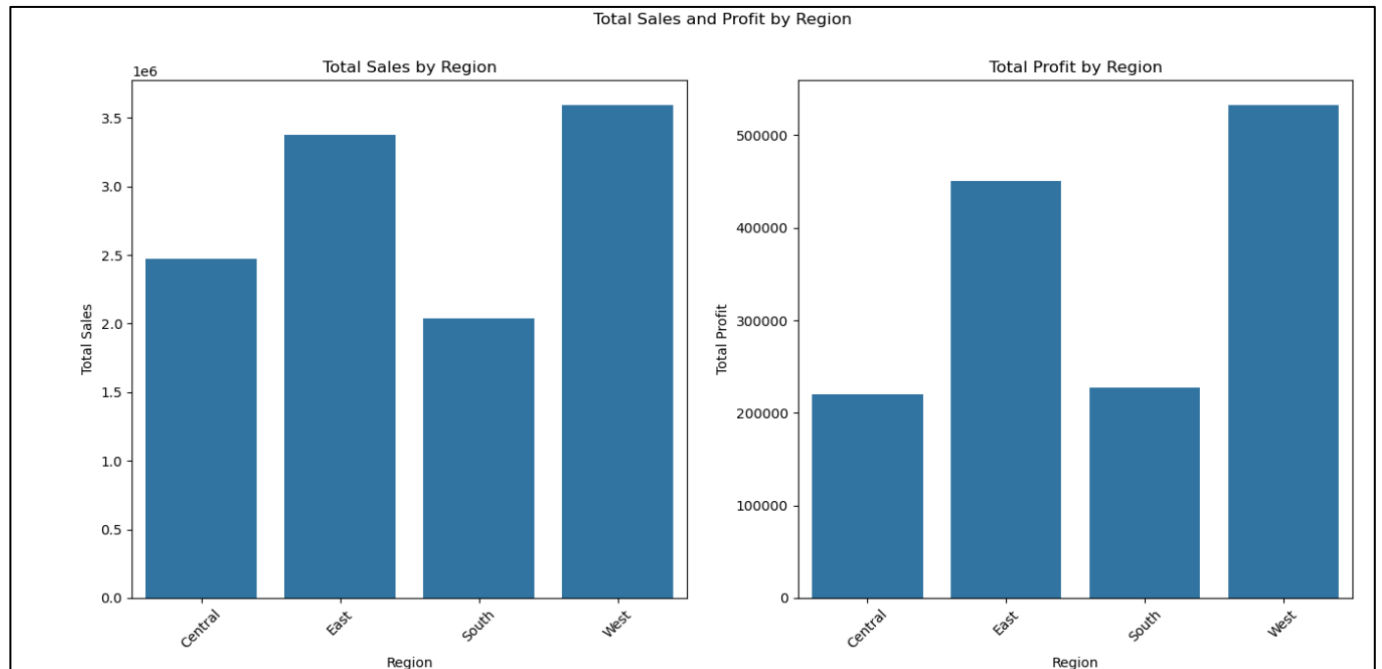
- **Data Cleaning:** Handled incorrect data entries (like negative sales prices), corrected inconsistent date formats, and removed 17 duplicate rows to ensure data integrity.
- **Missing Value Imputation:** Missing values in the Ship Mode (98 rows) and Quantity (18 rows) columns were imputed using logical rules and median values, respectively.
- **Outlier Handling:** A function was created to remove extreme outliers from the Sales Price column using the 3*IQR rule, making the dataset (cleaned_df) more reliable for distributional analysis.
- **Feature Engineering:** Created new, more insightful columns such as Total Sales (Price × Quantity), Total Profit, original_price, day_to_ship, and Shipping_Urgency to facilitate deeper analysis.

4. Analysis & Key Findings

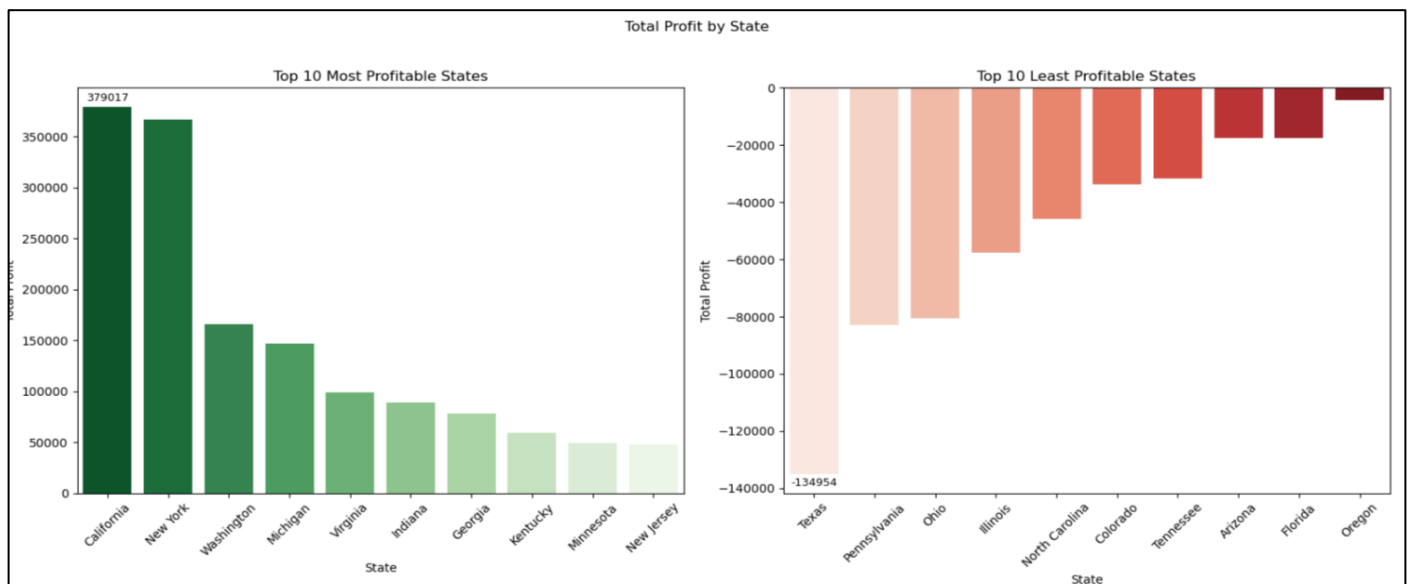
4.1 Geographical Performance: A Story of Two Extremes

Our analysis of geographical data shows a major imbalance.

- Regional Performance:** The West region is the undisputed leader in both sales and profitability. In contrast, the East region, despite having the second-highest sales, suffers from significantly lower profitability, indicating potential margin issues.



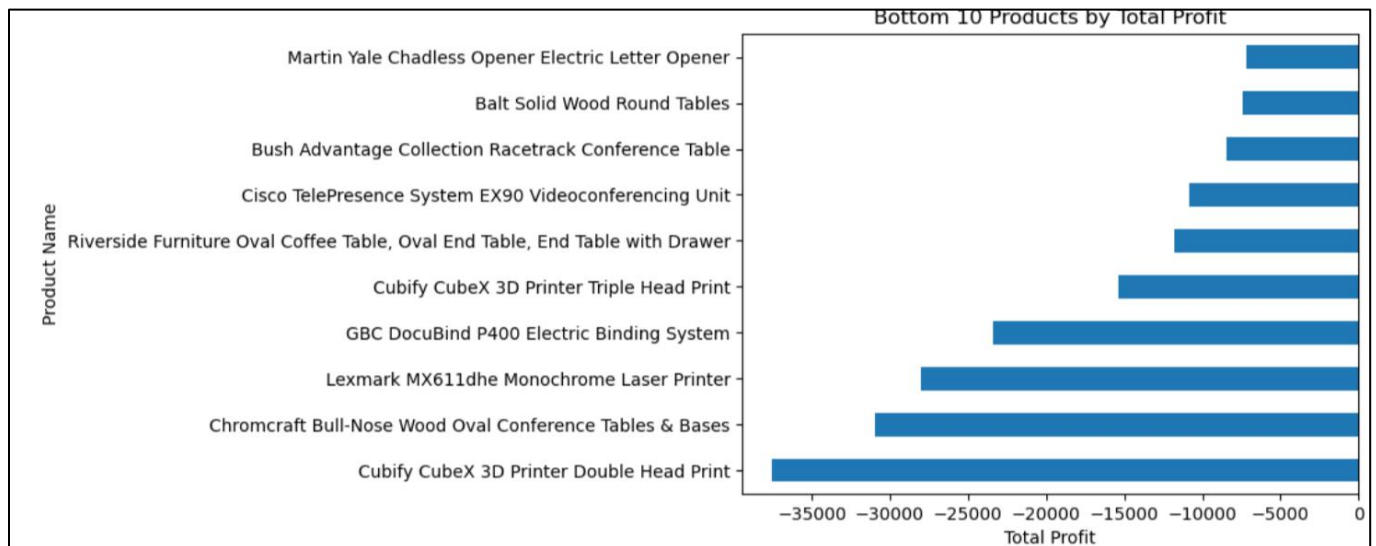
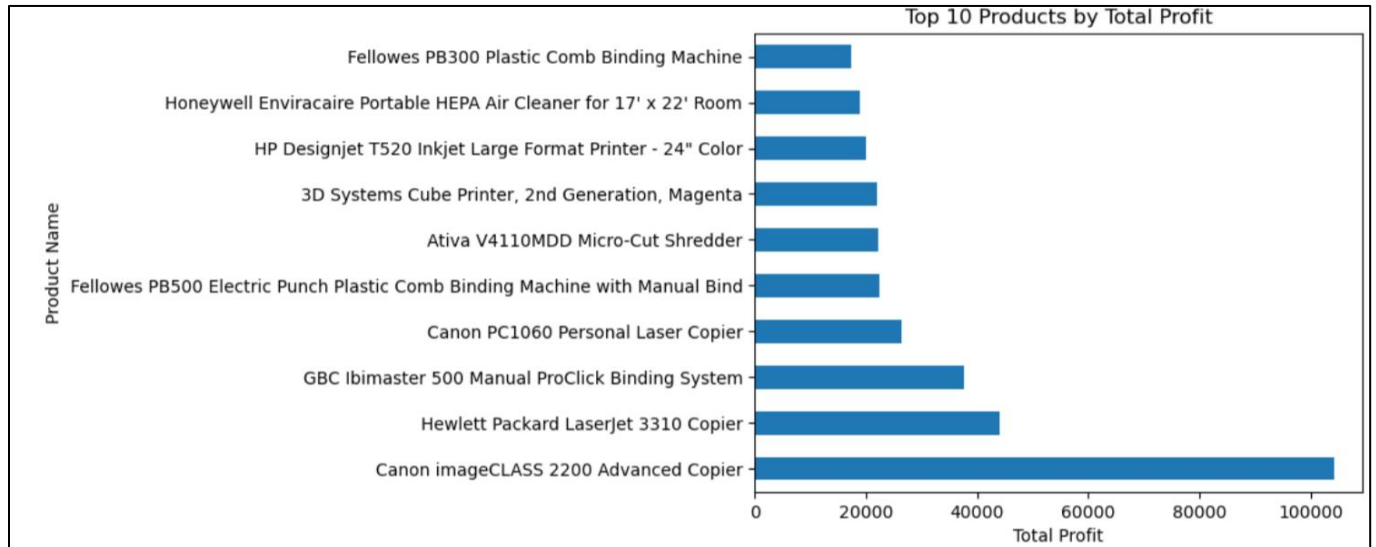
- State-Level Crisis:** At the state level, the situation is even more extreme. While **California** and **New York** are profitability powerhouses, states like **Texas**, **Pennsylvania**, and **Ohio** are massive loss centers, draining the company's resources.



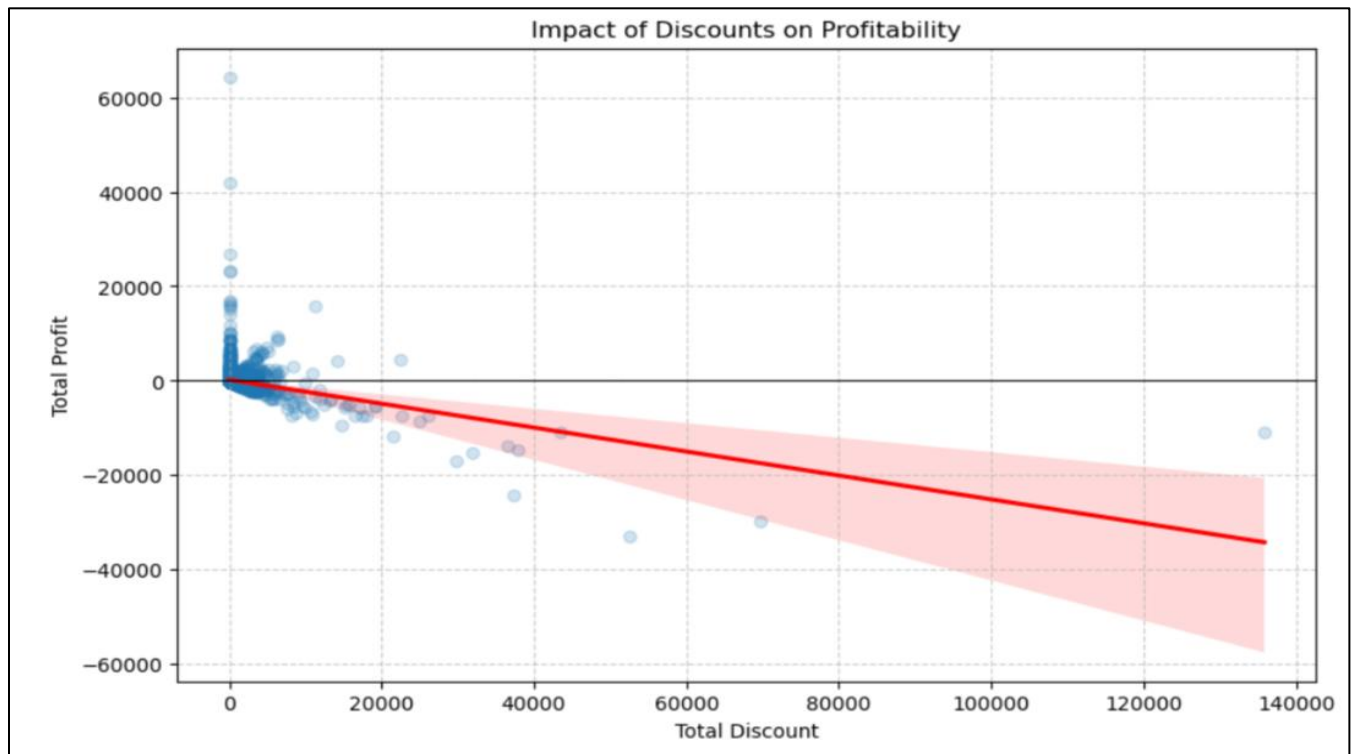
4.2 Profitability Drivers: Products and Discounts

Profit is driven by what we sell and how we sell it.

- Product Performance:** The company's profit is heavily reliant on a few high-value products, primarily advanced copiers and office machinery. Conversely, certain products, like the Cubify 3D printers, consistently result in significant losses.



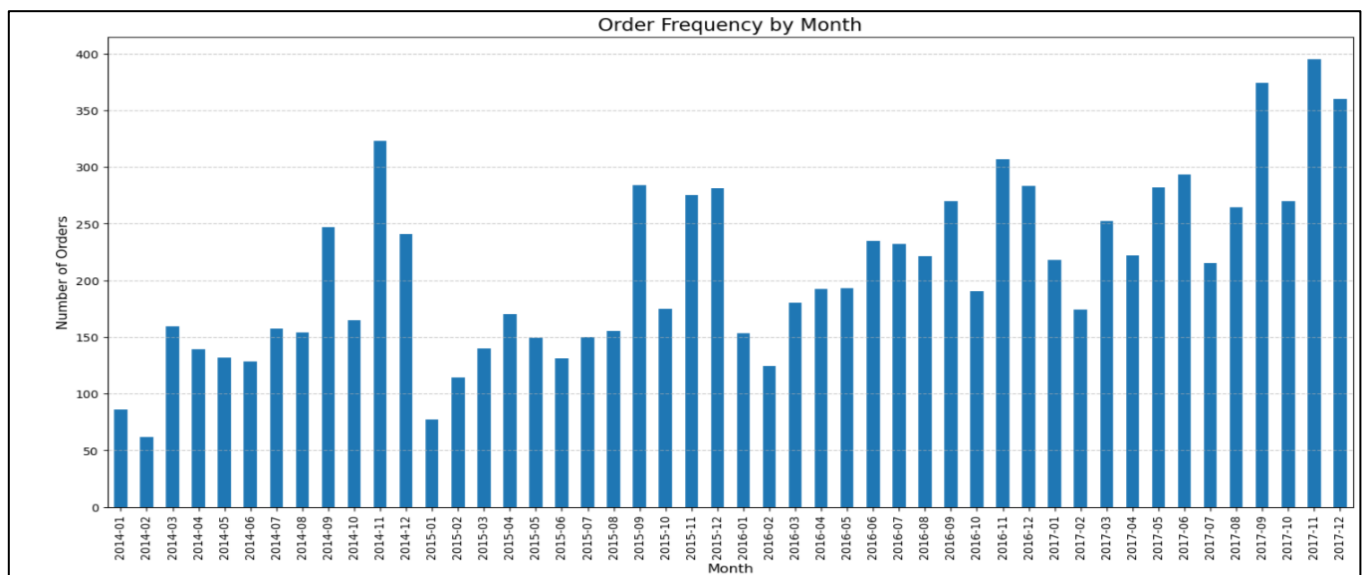
- **The Impact of Discounts:** The analysis clearly shows a strong negative correlation between discounts and profit. As discounts increase, profit consistently decreases. High discounts are the primary cause of unprofitable sales.



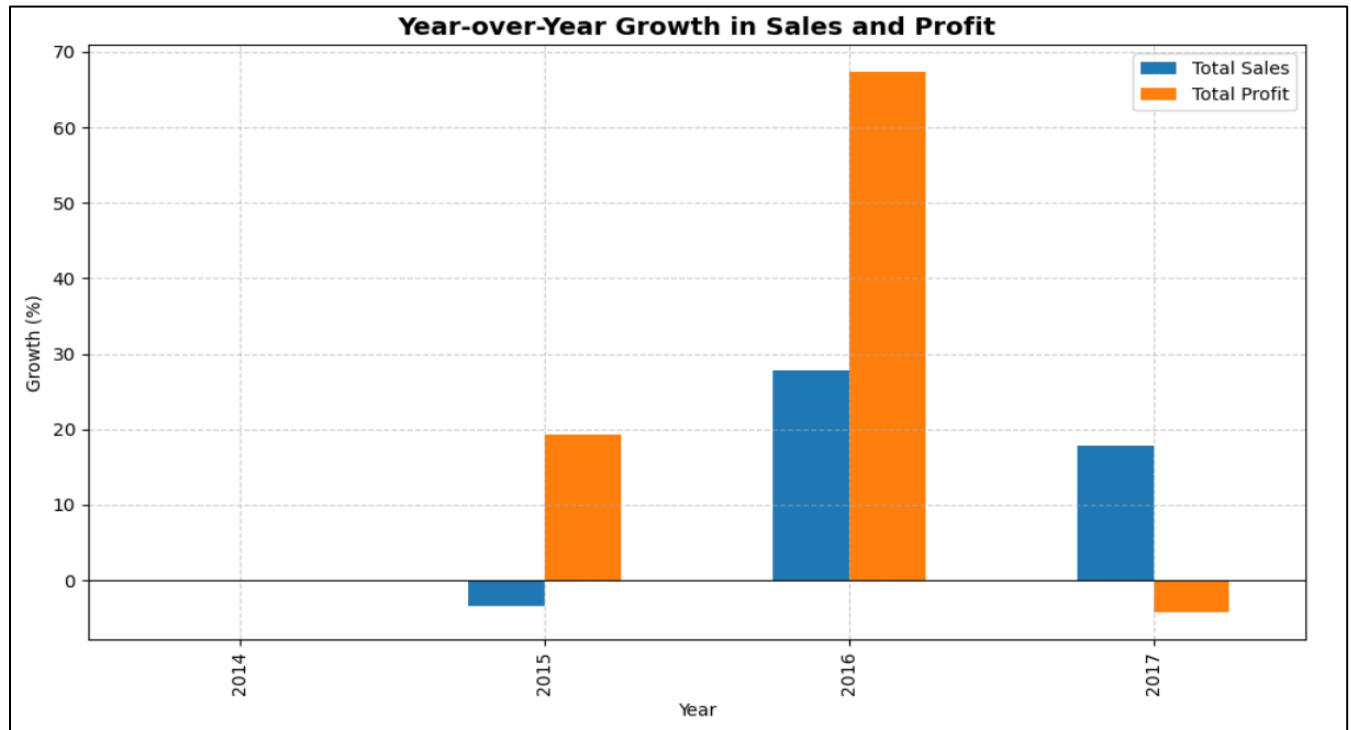
4.3 Temporal Patterns: Seasonality and Growth

The timing of sales reveals predictable patterns.

- **Seasonal Peaks and Dips:** The business operates on a strong seasonal cycle. Order volumes consistently peak in the last quarter (**September-December**) and hit a low in the first quarter (**January-February**) of each year.



- **Year-over-Year Growth:** The company shows a healthy overall growth trend. However, this growth is volatile. **2016** was an exceptional year where profit growth (nearly 70%) far outpaced sales growth (~30%). In contrast, **2017** was a red flag: sales grew by ~18%, but profits actually declined, indicating that the growth was unprofitable.



5. Recommendations

Based on the analysis, we recommend the following strategic actions:

1. Conduct a Regional Profitability Audit:

- **Action:** Immediately investigate the low profit margins in the **East** region. Analyze the product mix, shipping costs, and discount strategies specific to that region.
- **Action:** Form a task force to create a turnaround plan for **Texas, Pennsylvania, and Ohio**. Identify the root causes of the heavy losses and take immediate corrective action.

2. Leverage Seasonality for Maximum Gain:

- **Action:** Begin holiday marketing campaigns in late Q3 to capitalize on the predictable Q4 sales peak.
- **Action:** Optimize inventory for top-selling products by August to prepare for the year-end rush.
- **Action:** Launch a "New Year, New Office" promotion in January/February to counter the post-holiday slump.

3. Implement a Smart Discounting Strategy:

- **Action:** Review and cap the discount percentages for product categories that are already low-margin.
- **Action:** Completely stop offering discounts on products that are already identified as loss-making.

****End the Report****